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Impact of Urbanization on Local Hydrology: A Computational Approach for Stormwater Risk Assessment in Dhanmondi, Dhaka

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Introduction

Rapid urbanization increases impervious surfaces such as roads and buildings, reducing infiltration and increasing stormwater runoff. In cities like Dhaka, this leads to frequent urban flooding during heavy rainfall events. This project develops a computational urban flood prediction system to estimate runoff, flood depth, and drainage performance using hydrologic and hydraulic modeling techniques using MATLAB. The system also provides a user-friendly GUI for rapid flood risk assessment and urban drainage planning.

Background

- Mostly designed for large watersheds and limited tools for small urban areas.
- Lack of user-friendly flood assessment systems.
- Insufficient integrated urban drainage modeling.

Methodology

INPUT DATA
(RAINFALL INTENSITY, DURATION, LAND USE, PIPE DIAMETER)

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RUNOFF CALCULATION
USING RATIONAL METHOD

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PIPE FLOW ANALYSIS
USING MANNING'S EQUATION

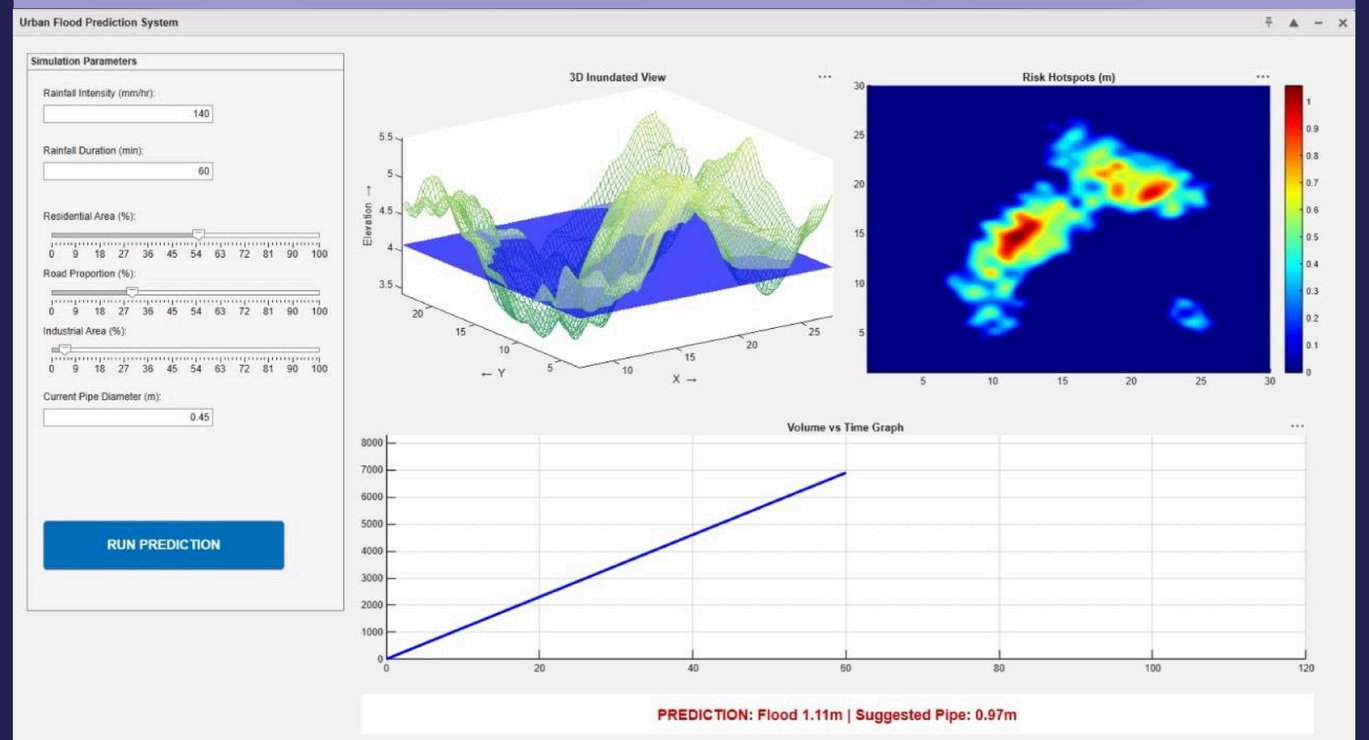
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FLOOD DEPTH MODELING
GRID-BASED NUMERICAL SIMULATION

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OUTPUT GENERATION
- PEAK RUNOFF AND FLOOD DEPTH
- STAGNANT WATER VOLUME
- RECOMMENDED PIPE DIAMETER

Findings



Results and Discussions

Key Findings

- Increasing impervious area significantly increases runoff.
- Existing drainage pipe capacity may become insufficient during intense rainfall.
- Flood depth increases rapidly under high rainfall intensity.
- Pipe diameter upgrades reduce overflow risk.

Conclusion

Urbanization significantly increases stormwater runoff and urban flood risk. The developed computational framework successfully predicts runoff, flood depth, and drainage performance, providing a simple and efficient tool for urban flood risk assessment and drainage planning.

References

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